

Broken Proportionality: Prosecutorial Demands and State Financial Loss in Indonesian Corruption Cases

Abstract

Does the punishment fit the crime in corruption cases? We estimate the elasticity of prosecution demands (*tuntutan*) with respect to state financial loss (*kerugian negara*) using 290 Indonesian Supreme Court corruption verdicts. The elasticity is 0.126 (bootstrap 95% CI [0.102, 0.156]): a 100-fold increase in state loss produces only a 1.8-fold increase in the prosecution demand. A Rp 10 million case draws a demand of 2.9 years; a Rp 100 billion case — ten thousand times the harm — draws only 9.2 years. Because punishment operates on a bounded scale while loss spans orders of magnitude, *some* compression is mathematically inevitable; the right benchmark is therefore comparative. Indonesia’s elasticity is roughly half that implied by the explicitly loss-graduated US Federal Sentencing Guidelines (about 0.25–0.30) and is reached with no guidelines at all — loss explains under 30% of demand variation ($R^2=0.285$). Prosecution demands are also less predictable than sentences ($R^2=0.315$ vs 0.600), meaning prosecutors exercise wider discretion than judges. Demands have risen over time (from roughly 5 years in 2014 to 7.5 years in 2025) while the sentencing discount has held steady at about 22%. Geographic variation disappears after controlling for case magnitude. The compressed elasticity means the marginal expected punishment for escalating corruption is negligible — the system grants an implicit volume discount on large-scale theft. Because judges anchor faithfully to these demands, the proportionality failure originates upstream, in prosecutorial discretion, not in judicial sentencing.

1. Introduction

Punishment should fit the crime. In corruption cases, “the crime” has a clear metric: the state financial loss (*kerugian negara*). A prosecutor deciding how many years to demand for a defendant who stole Rp 10 million faces a qualitatively different case from one involving Rp 100 billion. The question is whether prosecutors actually treat them differently — and if so, by how much.

Sentencing research overwhelmingly studies judges, treating prosecution demands as a given (Author, 2026b; Medvedeva et al., 2020; Strickson & De La Iglesia, 2020). This is a blind spot. Prosecution demands set the anchor from which judges work: in Indonesian corruption cases, prosecution demand alone explains 60% of sentencing variance (Author, 2026b). If the anchor itself is not proportional to the crime, judicial proportionality is impossible regardless of how carefully judges deliberate.

We test proportionality directly. Using 290 Indonesian Supreme Court corruption verdicts with documented state losses spanning seven orders of magnitude, we estimate the elasticity of prosecution demand with respect to *kerugian negara*. Indonesia is a useful setting because prosecutors operate without sentencing guidelines and the statutory range (1-20 years) is wide enough to accommodate the graduation observed in comparable systems.

The elasticity is 0.126. A Rp 100 billion case — ten thousand times the harm of a Rp 10 million case — draws a prosecution demand only 3.2 times longer. Some compression is unavoidable because punishment is bounded while loss is not; but as we show through comparison with the explicitly loss-graduated US sentencing system, Indonesia’s compression is roughly twice as severe and entirely unstructured. The consequences run in several directions: small-scale corruption is punished harshly per rupiah stolen, mega-corruption receives an implicit volume discount, and the marginal cost of escalating corruption is close to zero. The problem sits upstream of the judge, in

prosecutorial discretion — which means judicial reform alone cannot fix it.

Research Questions

- **RQ1:** Is prosecutorial demand proportional to state financial loss? What is the elasticity of demand with respect to loss magnitude?
- **RQ2:** What explains variation in prosecution demands beyond state financial loss? How predictable are prosecution demands from available case features?
- **RQ3:** Are there temporal or geographic patterns in prosecutorial severity?

2. Theoretical Framework

2.1 The Proportionality Principle

The proportionality principle — that punishment severity should reflect crime severity — is a foundational concept in sentencing theory (von Hirsch, 1992; Ashworth, 2015). In corruption cases, state financial loss provides an unusually clear measure of harm: unlike violent crimes where severity is multi-dimensional, the economic damage from corruption is quantifiable in monetary terms. If any offense type should exhibit proportional punishment, it is corruption.

What benchmark should we hold prosecutorial demands to? A naive standard would be an elasticity of 1.0 — a tenfold increase in loss yielding a tenfold increase in demand. But this is the wrong benchmark, and we reject it explicitly. Punishment operates on a bounded scale (the statutory range is 1–20 years; in practice almost all demands fall between 2 and 12 years), while corruption losses in our data span more than seven orders of magnitude. No bounded punishment scale can be log-proportional (elasticity 1.0) across such a range; substantial compression is a mathematical necessity, not a policy failure. An elasticity well below 1.0 is therefore expected in *any* sentencing system.

The meaningful question is comparative: how steeply do systems that *deliberately* tie punishment to loss actually graduate it, and where does Indonesia fall relative to that? The paradigm case is the US Federal Sentencing Guidelines §2B1.1, whose 16-tier “loss table” explicitly maps state/victim loss to offense-level increases (United States Sentencing Commission, 2024). Even this explicitly loss-graduated regime implies an elasticity on the order of 0.25–0.30: across the table, an approximately 3,800-fold increase in loss (from roughly USD 6,500 to USD 25 million) raises the offense level by about 20 levels, which corresponds to roughly a tenfold increase in sentence length. Notably, the US system is itself under active reform: in December 2025 the Sentencing Commission proposed collapsing the 16-tier table to 8 tiers precisely because of concerns that it over-weights loss and produces inconsistent, sometimes disproportionate outcomes (United States Sentencing Commission, 2025). The US thus represents a high-water mark of loss-based graduation — and even it is being pulled back. Against this benchmark, an Indonesian elasticity near 0.13 — roughly half the US figure, achieved with no guidelines whatsoever — indicates compression at an extreme of the observed cross-national range, reached not by design but as an emergent property of unstructured discretion.

2.2 Deterrence and Marginal Cost

Economic deterrence theory (Becker, 1968) models criminal behavior as a rational choice: individuals commit crimes when the expected benefit exceeds the expected cost (probability of detection

multiplied by punishment severity). For deterrence to function, the expected punishment must increase with the scale of the offense.

If prosecution demands are inelastic to state loss, the marginal expected punishment for additional corruption is near zero. A corrupt official deciding between diverting Rp 10 million and Rp 100 billion faces essentially the same prosecutorial demand — rendering the cost-benefit calculus heavily favorable toward larger-scale corruption. This creates a perverse incentive structure: the system punishes petty corruption relatively harshly while offering implicit leniency for mega-corruption.

2.3 Prosecutorial Anchoring

The anchoring literature (Tversky & Kahneman, 1974; Englich et al., 2006) suggests that decision-makers anchor to reference points and adjust insufficiently. Prosecutors may be anchored to a “normal” corruption demand range — an internalized sense of what constitutes an appropriate demand for corruption cases — and adjust minimally for case-specific factors including loss magnitude.

In a companion study (Author, 2026c), we documented bidirectional anchoring correction in judicial sentencing: judges pull extreme prosecution demands toward an implicit norm. The present study tests whether a parallel phenomenon exists on the prosecutorial side: do prosecutors compress their demands toward a central range regardless of case severity?

2.4 Prosecutorial Discretion

Prosecutors exercise enormous discretion in determining sentencing demands, yet this discretion receives far less scholarly attention than judicial discretion (Davis, 1969; Vorenberg, 1981). In Indonesia, no formal sentencing guidelines constrain prosecutorial demands. The Anti-Corruption Law (No. 31/1999 as amended by No. 20/2001) specifies statutory ranges — Pasal 2 (enrichment): 4-20 years; Pasal 3 (authority abuse): 1-20 years — but provides no formula linking loss magnitude to demand severity. This unstructured discretion creates the conditions for disproportionate outcomes.

3. Data and Methods

3.1 Corpus

We use *CorpusKorupsi* (Author, 2026a), a structured dataset of Indonesian Supreme Court corruption verdicts. From 693 scraped verdicts, 291 have both valid prosecution demands ($tuntutan > 0$) and documented state financial losses ($kerugian\ negara > 0$). Two data-quality issues affect the tails of the loss distribution and therefore the leverage of a log-log regression. First, a single case carries an implausible loss of Rp 100, almost certainly a parsing error (the next-smallest value is Rp 15 million); we drop it, yielding $n=290$ for the primary estimate. Second, one mega-case (the PT Timah tin-mining case, with an officially cited state loss of approximately Rp 300 trillion) appears three times because its case-level loss is attributed identically to three co-defendants — non-independent observations that we de-duplicate in a robustness check (Section 4.1). After removing the parsing error, losses range from Rp 15 million to Rp 300 trillion (median Rp 1.27 billion), spanning over seven orders of magnitude. As Section 4.1 shows, both corrections were slightly *deflating* the estimated elasticity; cleaning strengthens, rather than weakens, the central finding.

3.2 Variables

Variable	Description	n	Coverage
tuntutan_years	Prosecution demand (years)	291	100%
kerugian_negara	State financial loss (IDR)	291	100%
vonis_years	Sentence (years)	291	100%
has_pasal_2	Pasal 2 in judicial reasoning	291	100%
has_pasal_3	Pasal 3 in judicial reasoning	291	100%
tahun	Year of MA decision	291	100%
daerah	Court region	291	100%
discount	vonis / tuntutan ratio	291	100%

3.3 Statistical Methods

Proportionality test (RQ1). We estimate the elasticity of prosecution demand with respect to state financial loss using a log-log regression: $\log(\text{tuntutan}) = a + b * \log(\text{kerugian})$. Under proportionality, $b = 1.0$. We test whether the estimated elasticity is significantly less than 1.0.

Demand predictability (RQ2). We model prosecution demand using OLS regression with state financial loss, charge type (Pasal 2, Pasal 3), and year as predictors. We compare the R-squared with the demand-sentence relationship (R-squared = 0.60 from Author, 2026b) to assess relative predictability.

Temporal and geographic patterns (RQ3). Temporal trends are assessed using Spearman rank correlation. Geographic variation is tested using Kruskal-Wallis on both raw and residual (controlling for loss magnitude) demands.

4. Results

4.1 Prosecution Demands Are Highly Inelastic to State Loss (RQ1)

The log-log elasticity of prosecution demand with respect to state financial loss is **0.126** (bootstrap 95% CI [0.102, 0.156], $p < 0.001$, $R^2 = 0.285$) on the cleaned sample ($n = 290$). The entire confidence interval lies below the elasticity implied by the US loss table (about 0.25–0.30; Section 2.1), so even our upper bound does not reach the compression of the most explicitly loss-graduated system. Prosecution demands respond only weakly to the magnitude of harm.

In practical terms:

State financial loss	Predicted demand	Ratio to baseline
Rp 10 million	2.9 years	1.0x (baseline)
Rp 100 million	3.9 years	1.3x
Rp 1 billion	5.2 years	1.8x
Rp 10 billion	6.9 years	2.4x
Rp 100 billion	9.2 years	3.2x

A Rp 100 billion corruption case — involving state loss 10,000 times greater than a Rp 10 million case — receives a prosecution demand only 3.2 times higher. Each additional order of magnitude of loss multiplies the demand by only about 1.34 (a 34% increase per tenfold rise in harm).

Robustness. The estimate is stable across data-cleaning choices, and the uncleaned tail points were deflating rather than inflating it:

Specification	n	Elasticity	R ²
Full (uncleaned)	291	0.109	0.244
Drop parsing error (<Rp 1 juta) — <i>primary</i>	290	0.126	0.285
+ de-duplicate mega-case co-defendants	288	0.131	0.277
Winsorized at 1st–99th percentile	285	0.131	0.276

The low elasticity is also not an artifact of statutory constraints. The statutory maximum (20 years) binds only the very largest cases, and the statutory minimum for Pasal 2 (4 years) works *against* our finding: it forces small-case demands upward, inflating the apparent elasticity. Excluding cases with demands below 4 years (n=237) drops the elasticity further to 0.082 — compression is even more severe without the statutory floor. Controlling for whether the verdict mentions restitution (*pengembalian kerugian*, present in 35% of cases) does not change the elasticity (restitution coefficient about 0.25).

4.2 Prosecution Demands Are Less Predictable Than Sentences (RQ2)

A model including state financial loss, charge type, and year explains only **31.5%** of variation in prosecution demands (R-squared=0.315). This is substantially lower than the 60% of sentencing variance explained by prosecution demand alone (Author, 2026b).

Model	R-squared
tuntutan ~ log(kerugian)	0.297
tuntutan ~ log(kerugian) + Pasal 2 + Pasal 3	0.315
tuntutan ~ log(kerugian) + Pasal 2 + Pasal 3 + year	0.321
vonis ~ tuntutan (Paper 2 benchmark)	0.600

Adding charge type improves the model marginally: Pasal 2 cases receive demands 1.1 years higher than equivalent cases (p=0.019). Pasal 3 is not independently significant (p=0.109). Year adds no significant explanatory power (p=0.113).

Put differently: judges are well-anchored to what prosecutors ask for (R-squared=0.60), but prosecutors themselves are only loosely tethered to the objective severity of the case. The discretion gap lies upstream.

4.3 Temporal Increase in Prosecutorial Severity (RQ3)

Prosecution demands have increased significantly over the observation period (Spearman rho=0.773, p=0.005), from approximately 5 years in 2014-2015 to approximately 7.5 years in 2025-2026. However, the sentencing discount (vonis/tuntutan) has not changed (Spearman rho=0.145, p=0.670), indicating that judges absorb the increased demands proportionally rather than resisting the upward trend.

One reading: as anti-corruption rhetoric has intensified, prosecutors have raised their demands — but judges keep cutting by the same fraction, so the actual sentences have risen more slowly than the demands.

4.4 Geographic Variation Is a Composition Effect

Raw prosecution demands vary significantly across provinces (Kruskal-Wallis $H=40.0$, $p<0.001$). However, after controlling for state financial loss, the geographic variation disappears ($H=22.2$, $p=0.104$). Regions with higher average demands (DKI Jakarta, Sumatera Selatan) handle larger-magnitude cases, not systematically harsher prosecutors.

The same result appears in judicial sentencing (Author, 2026b): what looks like regional disparity is really case mix. Prosecutors in different regions are not systematically harsher or softer — they handle different kinds of cases.

5. Discussion

5.1 Why Proportionality Fails

An obvious objection is that state financial loss is only one dimension of crime severity. Prosecutors legitimately consider the defendant’s position, cooperation, modus operandi, and restitution when calibrating demands. A low elasticity with respect to *kerugian* alone does not necessarily indicate broken proportionality if other factors compensate. We address this in two ways. First, controlling for the available multi-dimensional indicators — charge type (Pasal 2 vs 3), restitution mentions, and region — does not change the elasticity. Second, even granting that severity is multi-dimensional, an elasticity of 0.126 is extreme relative to the comparative benchmark: it implies that a case involving ten thousand times more public harm receives only about three times the punishment demand, roughly half the graduation built into the US loss table. No plausible weighting of non-monetary severity factors can close a gap this large — and such factors would have to be *systematically anticorrelated* with loss to do so.

Three mechanisms may explain the compression:

Anchoring to norms. Prosecutors may anchor to an implicit “normal” corruption demand — approximately 5-8 years — and adjust insufficiently for case severity. This parallels the judicial anchoring documented in companion studies (Author, 2026b, 2026c). The statutory range of 1-20 years is broad enough to accommodate proportional demands, but prosecutors appear to use only a narrow band within this range.

Evidentiary complexity. Larger corruption cases typically involve more complex financial structures, making it more difficult to establish the full extent of state loss. Prosecutors may discount their demands to reflect evidentiary uncertainty, even when the documented loss is large.

Defendant power asymmetry. Mega-corruption cases typically involve powerful defendants — governors, ministers, directors of state enterprises — who can mount more aggressive legal defenses. Prosecutors may temper their demands in anticipation of more contested proceedings, a form of strategic adjustment documented in other contexts (Bibas, 2004).

5.2 Implications for Deterrence

If the expected additional punishment for increasing the scale of corruption from Rp 10 million to Rp 100 billion is only 6.3 years, the marginal cost of additional corruption is approximately Rp 16 billion per year of additional imprisonment. For a rational actor, this makes larger-scale corruption overwhelmingly favorable: the “price” of corruption, measured in prison time per rupiah stolen, decreases by approximately 99.97% as the scale increases from Rp 10 million to Rp 100 billion.

Within a rational-choice framework, this broken proportionality predicts weak deterrence against escalation: once a public official has crossed into corruption, the marginal punishment for increasing the scale is negligible. We do not claim that all corruption is rationally calculated — institutional culture, opportunity structures, and social norms all play roles (Lambsdorff, 2007). But to the extent that punishment signals matter at all, the current signal is perverse: the system imposes a near-flat tariff regardless of scale.

5.3 Prosecutorial Discretion as the Upstream Problem

Our finding that prosecution demands are less predictable than sentences ($R^2=0.315$ vs 0.600) shifts the focus of the proportionality problem upstream. The “judicial opacity” identified in Author (2026b) — where 40% of sentencing variance is unexplained — may be partially inherited from prosecutorial opacity. Judges are anchored to demands that are themselves not proportional to case severity.

Cross-national comparisons support this interpretation. Our sentence-from-demand R^2 of 0.60 matches the Dutch sentencing benchmark (van Wingerden et al., 2016, $R^2=0.60$ with multiple predictors) and exceeds the majority of 28 sentencing regression studies reviewed by Wittenbrink and Niehaus (2022). The approximately 40% unexplained sentencing variance is thus not a distinctively Indonesian phenomenon — it is consistent with judicial discretion levels observed globally.

What IS distinctive is the *degree* of compression at the prosecutorial stage. The demand-loss elasticity of 0.126 is roughly half the about 0.25–0.30 implied by the US loss table — the most explicitly loss-graduated regime in comparative practice — yet Indonesia reaches this compression with no sentencing or charging guidelines at all. Where the US graduates punishment by loss through a deliberate, contested, and now-reforming 16-tier schedule (United States Sentencing Commission, 2024, 2025), Indonesian prosecutors compress the vast range of corruption severity into a narrow demand band as an unstructured by-product of discretion. That band then propagates through the faithfully anchored judicial system, so a reform aimed only at judges would leave the binding constraint untouched.

5.4 Policy Implications

Prosecutorial guidelines. Formal guidelines linking prosecution demand to state financial loss — similar to US Sentencing Guidelines’ loss tables — could restore proportionality. Such guidelines need not eliminate prosecutorial discretion entirely; even a requirement to justify deviations from a proportional baseline would increase transparency.

Graduated prosecution. The current flat demand structure could be replaced with a graduated system where demands increase more steeply with loss magnitude. The existing statutory ranges (1-20 years) are sufficient to accommodate proportional demands.

Transparency. Requiring prosecutors to document the relationship between their demands and case severity measures would create accountability without constraining discretion. Currently, the demand-setting process is opaque and unstructured.

5.5 Limitations

Selection bias. Our corpus consists of Supreme Court cassation decisions — appealed cases. The demand-loss elasticity may differ at the district court level, where the full population of corruption

cases is processed. If cases with extreme disproportionality are more likely to be appealed, our estimates may not generalize.

Kerugian negara measurement. State financial loss figures are extracted from verdict metadata and may not reflect the actual harm. Loss quantification in corruption cases is contested and sometimes reflects prosecutorial framing rather than objective measurement.

Confounders. Variables not captured in our data — defendant cooperation, plea agreements, strength of evidence, defendant status, case complexity — may explain some of the demand variation we attribute to prosecutorial discretion. However, even if these confounders partially explain the low elasticity, the finding that $R^2=0.285$ for the loss-demand relationship remains consequential: loss magnitude is a weak predictor of prosecution demand regardless of mechanism.

Tuntutan level. Our tuntutan variable may refer to the original district court demand or to the prosecution’s position at cassation. Two observations suggest it is the original demand. First, the mean sentencing discount ($\text{vonis}/\text{tuntutan} = 0.78$) implies judges consistently reduce demands by ~22% — a pattern consistent with PN-level demands being discounted, not with cassation-level demands that would have been adjusted to approximate the expected outcome. Second, in 75% of cases vonis falls below tuntutan, inconsistent with cassation-level demands that would more closely track the eventual sentence. Nevertheless, we cannot rule out that some tuntutan values reflect cassation-stage adjustments, which would compress the demand range and bias our elasticity estimate downward.

6. Conclusion

We document a striking failure of proportionality in Indonesian corruption prosecution: the elasticity of prosecution demand with respect to state financial loss is only 0.126, meaning that a 10,000-fold increase in corruption scale produces only a 3.2-fold increase in the prosecution demand. Because punishment is bounded, some compression is inevitable; but Indonesia’s elasticity is roughly half that of the explicitly loss-graduated US sentencing system and is achieved without any guideline structure. This inelasticity is not explained by statutory caps, geographic variation, or charge type.

The broken proportionality creates a regressive prosecution structure where small-scale corruption is punished disproportionately harshly relative to harm, while mega-corruption receives an implicit “volume discount.” This has direct implications for deterrence: the marginal punishment for escalating corruption is negligible, creating a rational-actor incentive to maximize the scale of corrupt acts.

Our findings shift the focus of corruption sentencing reform from judges to prosecutors. Judicial sentencing in Indonesia is well-anchored to prosecution demands ($R\text{-squared}=0.60$, consistent with cross-national benchmarks). The proportionality failure originates upstream, in prosecutorial demand-setting, where only 31.5% of variation can be explained by available case features. Restoring proportionality requires addressing prosecutorial discretion — through guidelines, transparency requirements, or graduated demand structures — not merely reforming judicial sentencing.

The CorpusKorupsi dataset and analysis code are publicly available at [repository URL].

Declarations

Funding. This research received no external funding. **Conflicts of interest.** The author declares no conflicts of interest. **Ethics approval.** This study analyzes publicly available court documents published by the Indonesian Supreme Court. No human subjects were involved and no ethics approval was required. **Data availability.** The CorpusKorupsi structured dataset and analysis scripts will be made available upon publication. **Use of AI-assisted tools.** The author used Claude (Anthropic, Claude Opus) as a computational research assistant for programming, statistical analysis, and manuscript drafting. All analyses were independently verified through reproducible scripts, and the author takes full responsibility for all scientific claims and interpretations.

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